



DRISHTI

ENTERPRISE API REFERENCE

Drishti — API Reference Manual

Complete REST Endpoints, Payloads, Schemas & Authentication



Graph-Powered Attack Paths



Live Watch Agent



Dollar Risk Quantifier



AI Remediation Guardrails



URL Trust Scoring

PLATFORM VERSION

v0.2 Enterprise Edition

RELEASE DATE

August 2026

ARCHITECTURE TIER

Full-Stack Multi-Tenant



RESTRICTED // INTERNAL ARCHITECTURE & TECHNICAL SPECIFICATION // FOR AUTHORIZED USE ONLY

Drishti — API Reference

Reverse-engineered from the implemented FastAPI backend. All 14 routers, every endpoint, request/response schemas, authentication, rate limits, and error envelope — organized by route group.

Last updated: 2026-08-17 — verified against source code.

1. Error envelope

All errors (except auth middleware) return:

JSON

```
{
  "error": {
    "type": "string",
    "title": "Human-readable summary",
    "status": 400,
    "detail": "Detailed message",
    "instance": "Optional context"
  }
}
```

Status codes: `standard_http` (400, 401, 403, 404, 422, 409, 413, 429, 500).

2. Authentication

Header	Value	Used by
<code>Authorization: Bearer <access_token></code>	JWT (HS256)	User endpoints
<code>Authorization: Bearer drishti_<base64></code>	Hashed agent token	Agent ingest

Refresh: `POST /api/auth/refresh` with `{refresh_token}`.

3. Route catalog

3.1 Health

Method	Path	Auth	Purpose
GET	/	None	Liveness probe
GET	/health	None	Detailed health

3.2 Auth (/api/auth)

Method	Path	Auth	Rate Limit	Purpose
POST	/register	None	90/min (burst 60)	Create org + admin user
POST	/login	None	90/min (burst 60)	Issue access + refresh tokens
POST	/refresh	None	—	Rotate access token
POST	/logout	User	—	Revoke refresh token
GET	/me	User	—	Current user profile

Request schemas:

YAML
<pre>RegisterRequest: name: string email: string password: string # min 8 chars org_name: string LoginRequest: email: string password: string RefreshRequest: refresh_token: string</pre>

Response:

YAML

```
TokenResponse:
  access_token: string
  refresh_token: string
  token_type: "bearer"
```

3.3 Organization (/api)

Method	Path	Auth	Role	Purpose
GET	/org	User	—	Org info
POST	/org/load-sample	User	admin	Load Acme sample network
POST	/org/reset	User	admin	Drop all org data
POST	/org/agent-token	User	admin	Issue new agent token
DELETE	/org/agent-token	User	admin	Revoke agent token
GET	/org/members	User	admin	List org members

3.4 Ingest (/api)

Method	Path	Auth	Rate Limit	Purpose
POST	/ingest	Agent	60/min (burst 20)	Idempotent host/service/vuln upsert + recompute

Request:

YAML

```
IngestRequest:
  org_slug: string
  hostname: string
  ip: string
  asset_type: string # webapp, server, database, firewall, device, unknown
  services: IngestService[]
  vulnerabilities: IngestVulnerability[]
  connectivity: IngestConnection[]
  scan_metadata: IngestScanMetadata

IngestService:
  port: integer
  protocol: string # tcp, udp
  name: string
  version: string
  is_open: boolean

IngestVulnerability:
  id: string # e.g., CVE-2024-1234
  title: string
  severity: string
  cvss: float
  description: string

IngestConnection:
  from: string # IP
  to: string # IP
  relation: string # network, admin, trust, exposure
  via_service?: string
  via_cve?: string
```

Response:

YAML

```
IngestResponse:
  asset_id: string
  services_upserted: integer
  vulnerabilities_upserted: integer
  connections_upserted: integer
  findings_created: integer
  findings_resolved: integer
  recompute_triggered: boolean
```

3.5 Graph (/api)

Method	Path	Auth	Purpose
GET	/graph	User	Attack map data (React Flow)
GET	/stats	User	Engine stats (recompute count, ms, AI counts)

Response (/graph):

YAML

```
GraphResponse:
  nodes: Node[]
  edges: Edge[]
  zones: ZoneInfo[]
  live_devices: NetworkDevice[]
  network_threats: NetworkThreat[]

Node:
  id: string
  type: "asset" | "gateway" | "device" | "threat"
  position: { x: float, y: float }
  data:
    label: string
    ip: string
    hostname: string
    zone: string
    risk_score: float
    asset_type: string
    criticality: string
    is_crown_jewel: boolean
    services: Service[]
    blast_radius_count: integer
    downstream_value_usd: float
    threat?: { kind, severity, mitre, recommendation }
    demo_label?: string

Edge:
  id: string
  source: string
  target: string
  label: string # port/cve/via_service
  risk_score: float
  onTopPath: boolean
  path_id?: string
  style:
    stroke: string
    strokeWidth: number
    animated: boolean
  data:
    relation: string
```

```
port?: number
cve?: string
via_service?: string
```

3.6 Paths (/api)

Method	Path	Auth	Purpose
GET	/paths	User	List ranked attack paths
GET	/paths/{path_id}	User	Path detail (hop-by-hop)
GET	/paths/{path_id}/blast-radius	User	Downstream assets + value

Response (/paths):

YAML

```
PathSummary:
  path_id: string
  entry: { ip, hostname }
  target: { ip, hostname, zone, risk_score }
  hops: integer
  risk_score: float # 0-100
  likelihood: float # 0-1
  impact_usd: float # engine-computed
  narrative: string # AI-generated or template
  top_hop_labels: string[]
  top_cves: string[]
```

3.7 Findings (/api)

Method	Path	Auth	Purpose
GET	/findings	User	List findings
PATCH	/findings/{id}	User	Update status

Query params: ?status=open|remediating|resolved|accepted&severity=critical|high|medium|low

Request (PATCH):

YAML

```
FindingUpdate:
  status: string # open, remediating, resolved, accepted
  accepted_until?: datetime
```

Response:

YAML

```
FindingOut:
  id: string
  asset_id: string
  asset_ip: string
  asset_hostname: string
  cve_id: string
  title: string
  severity: string
  cvss: float
  port: integer
  service_name: string
  status: string
  auto_resolved: boolean
  accepted_until: string
```

3.8 Assets (/api)

Method	Path	Auth	Purpose
GET	/assets	User	List assets
GET	/assets/{id}	User	Asset detail
PATCH	/assets/{id}	User	Update asset

Response (`AssetSummary`):

YAML

```
AssetSummary:
  id: string
  ip: string
  hostname: string
  zone: string
  asset_type: string
  criticality: string
  internet_facing: boolean
  risk_score: float
  is_crown_jewel: boolean
  blast_radius_count: integer
  downstream_value_usd: float
  last_scanned_at: string
```

Response (`AssetDetail`):

YAML

```
AssetDetail:
  id: string
  ip: string
  hostname: string
  os: string
  zone: string
  asset_type: string
  criticality: string
  internet_facing: boolean
  base_value_usd: float
  risk_score: float
  is_crown_jewel: boolean
  blast_radius_count: integer
  downstream_value_usd: float
  services: ServiceOut[]
  findings: FindingOut[]
  hardening: NodeHardening[]
```

3.9 Dashboard (/api)

Method	Path	Auth	Purpose
GET	/dashboard	User	Dashboard data
GET	/stats	User	Engine + AI stats
POST	/recompute	User	Trigger manual recompute

Response (/dashboard):

YAML

```
Dashboard:
  total_exposure_usd: float
  open_findings: integer
  critical_assets: integer
  top_path: PathSummary
  paths: PathSummary[]
  zone_summary: ZoneSummary[]
  severity_counts: { critical: int, high: int, medium: int, low: int }
  last_recompute_at: string
  recompute_ms: float

ZoneSummary:
  zone: string
  count: integer
  exposure_usd: float
  avg_risk: float
  critical_count: integer
```

3.10 Report (/api)

Method	Path	Auth	Purpose
GET	/report/cves	User	All CVEs for org
GET	/report/distribution	User	Severity distribution chart
GET	/report/hardening	User	Per-node hardening projections
GET	/report/network-summary	User	Network summary
GET	/report/ml-summary	User	AI/ML call statistics

3.11 AI (/api/ai)

Method	Path	Auth	Rate Limit	Purpose
POST	/ai/remediate	User	20/min (burst 6)	Generate remediation
POST	/ai/impact	User	20/min	AI impact narrative
POST	/ai/predict	User	20/min	Forward-looking prediction
POST	/ai/network-summary	User	20/min	Executive summary
POST	/ai/block	User	20/min	Block a risky domain

Request (/remediate):

YAML

```
RemediateRequest:
  finding_id: string
  regenerate?: boolean # force new LLM call

RemediateResponse:
  id: string
  finding_id: string
  kind: string # ansible, shell, cloud_cli, manual
  title: string
  summary: string
  script: string
  steps: Step[]
  estimated_risk_reduction: float
  requires_restart: boolean
  reviewed: boolean
  disclaimer: string
  model: string
```

Refused response:

YAML

```
{
  "refused": true,
  "reason": "Request could not be answered defensively."
}
```

3.12 Live (/api)

Method	Path	Auth	Rate Limit	Purpose
GET	/live/devices	User	—	Live network devices
POST	/live/devices	Agent	60/min	Ingest device list
GET	/live/observe	User	—	Observed domains
POST	/live/observe	User/Agent	—	Add domain observation
POST	/live/sync-active	Agent	60/min	Active tabs/apps
GET	/live/network-threats	User	—	Detected threats
POST	/live/deep-scan	User	—	Consent-gated deep scan
POST	/live/deep-scan/scan-subnet	User	—	Full subnet scan
GET	/live/autoscan	User	—	Autoscan config
PUT	/live/autoscan	User	—	Update autoscan config
POST	/live/demo-attack	User	—	Inject demo attack
POST	/live/demo-attack/clear	User	—	Clear demo attack

Request (/live/deep-scan):

YAML

```
DeepScanRequest:
  ip: string
  cidr?: string
  ports?: string # e.g., "1-65535" or "22,80,443"
  consent: true # must be exactly True, not truthy
```

3.13 Netconfig (/api)

Method	Path	Auth	Purpose
POST	/netconfig/analyze	User	Run DMZ/NAT/DHCP detectors

Request:

YAML

```
NetconfigAnalyzeRequest:
  scan_dmz: boolean
  scan_nat: boolean
  scan_dhcp: boolean
```

Response:

YAML

```
NetconfigResult:
  dmz: { status: string, source: string, evidence: string }
  nat: { status: string, source: string, evidence: string }
  dhcp: { status: string, source: string, evidence: string }
```

3.14 URL Trust (/api/url-analyzer)

Method	Path	Auth	Purpose
POST	/url-analyzer/analyze	User/Agent	Analyze a URL
GET	/url-analyzer/history	User	Past analyses

Request:

YAML

```
URLAnalysisRequest:
  url: string
  observe_domains?: boolean # add to LiveObservations
```

Response:

YAML

```
URLAnalysisResult:
  url: string
  hostname: string
  score: integer # 0-100
  band: "trusted" | "caution" | "high_risk"
  signals: object # provider responses (never from cache)
  website: object # static site data
  providers: object # provider config
  summary: object # aggregated
```

3.15 Assets (/api)

Method	Path	Auth	Purpose
GET	/assets	User	List assets
GET	/assets/{id}	User	Asset detail
PATCH	/assets/{id}	User	Update asset

3.16 Remediation (frontend-triggered, server-routed)

Method	Path	Auth	Purpose
POST	/remediation/{finding_id}	User	Generate remediation
PATCH	/remediation/{id}	User	Update (reviewed)
POST	/remediation/{id}/regenerate	User	Force new LLM call

4. Rate limits (detail)

Bucket	Rate	Burst	Key
Auth (register/login/refresh)	90/min	60	<code>auth:<email_hash></code>
Ingest	60/min	20	<code>ingest:<agent_token_hash[:8]></code>
AI	20/min	6	<code>ai:<user_id></code>
Body size	—	1 MB hard cap	All requests

5. CORS

Source	Pattern
<code>http://localhost:5173</code>	Dev web server
<code>http://localhost:5174</code>	Dev fallback port
<code>chrome-extension://*</code>	Chrome extension (any ID via regex)

Allow credentials: yes. Allow methods: `GET, POST, PUT, DELETE, PATCH, OPTIONS`.

6. Middleware stack (order)

1. **MaxBodySizeMiddleware** — ASGI streaming body cap (1 MB)
2. **TimingMiddleware** — Adds `X-Process-Time` header
3. **RateLimitMiddleware** — Token bucket (per-group keying)
4. **AuthMiddleware** — JWT or agent-token parsing
5. **CORSMiddleware** — Allowlist

7. Endpoint groups with group-level auth

Group	Auth	Role
<code>ingest</code>	Agent token	<code>org_id</code> match
<code>auth</code>	None (self-auth)	—
<code>admin</code> (<code>/org/*</code>)	User JWT	<code>admin</code> role
<code>ai</code>	User JWT	—
<code>live</code>	User JWT or Agent token	<code>org_id</code> match
<code>netconfig</code>	User JWT	—
<code>url-analyzer</code>	User JWT	—
<code>graph</code> , <code>paths</code> , <code>findings</code> , <code>assets</code> , <code>dashboard</code> , <code>report</code>	User JWT	—